Expectation, Variance & Indicators: 20 Worked Problems

Linearity · Var. of Sums · Indicator Identity · Fixed Points · Runs · Inversions · Coupon Collector

This is arguably the single highest-leverage chapter in the whole syllabus: the Method of Indicator Variables turns "impossible to compute directly" problems (fixed points of a random permutation, inversions, coupon collecting) into trivial sums of easy pieces. Every problem below is color-coded by sub-topic and carries a plain analogy — because the real skill here isn't the algebra, it's learning to *see* a complicated random variable as a sum of simple 0/1 indicators in the first place.

I. Expectation & Variance Algebra

Problem A1 — Linearity needs no independence, no correlation knowledge, nothing

A desk runs 5 heavily correlated strategies (all driven by the same macro factor). Given $\mathbb{E}[P\&L_i] = \$2\text{k}, \$1.5\text{k}, \$3\text{k}, -\$0.5\text{k}, \$2.5\text{k}$, find $\mathbb{E}[\text{total P\&L}]$.

Solution. Linearity holds under *arbitrary* dependence — we don't need to know the correlation structure at all:

$$\mathbb{E}[\text{Total}] = 2 + 1.5 + 3 - 0.5 + 2.5 = \boxed{\$8.5\text{k}}.$$

★ *Intuition:* This is the most underrated fact in all of probability: linearity of expectation places *zero* requirements on independence. You can know literally nothing about how these 5 strategies move together and still answer this question with first-grade arithmetic. Every indicator-variable trick in this entire chapter is secretly just this one fact, applied to 0/1 random variables.

Problem A2 — Scaling and subtraction under dependence

Strategy X (mean \$500, volatile) and hedge Y (mean $-\$200$) are strongly, intricately correlated by design. The desk trades 3 units of X and 2 units of Y. Find $\mathbb{E}[3X - 2Y]$.

Solution.

$$\mathbb{E}[3X - 2Y] = 3\mathbb{E}[X] - 2\mathbb{E}[Y] = 3(500) - 2(-200)$$

$$= 1500 + 400 = \boxed{\$1{,}900}.$$

★ *Intuition:* Notice the minus sign on a *negative*-mean variable flips to a plus — subtracting a hedge you expect to lose money adds to your expected total, exactly as a hedge should. The arithmetic enforces correct intuition automatically, even though we still know nothing about $\text{Cov}(X, Y)$.

Problem A3 — Linearity through wild path-dependence

An HFT desk places 10 sequential orders with brutally complex path-dependence (each order's P&L depends on the entire history of prior orders in a way no one could write down in closed form). You're told only that $\mathbb{E}[P\&L_i] = (-1)^i \binom{10}{i}$ for $i = 1, \ldots, 10$. Find $\mathbb{E}[\text{Total}]$.

Solution. Path-dependence is irrelevant to linearity — sum the means directly:

$$\mathbb{E}[\text{Total}] = 10 \sum_{i=1}^{10} \frac{(-1)^i}{i} \approx \boxed{-\$6.46}.$$

★ *Intuition:* As the desk runs forever ($n \to \infty$), this alternating sum converges to $-10 \ln 2 \approx -\$6.93$ — the famous alternating harmonic series. The lesson is bigger than the number: no matter how tangled the dependence between terms, as long as you're only asked for a *total expectation*, linearity lets you ignore the tangle completely and just add up individual means.

Problem B1 — Variance of a sum: hedging flips the sign of the effect

Two strategies have $\text{Var}(X) = 16$, $\text{Var}(Y) = 9$, $\text{Cov}(X, Y) = -6$ (designed to hedge each other). Find $\text{Var}(X + Y)$ and $\text{Var}(X - Y)$.

Solution.

$$\text{Var}(X + Y) = 16 + 9 + 2(-6) = 25 - 12 = \boxed{13}.$$

$$\text{Var}(X - Y) = 16 + 9 - 2(-6) = 25 + 12 = \boxed{37}.$$

★ *Intuition:* The same negative covariance that makes the *sum* safer (13, below either individual variance) makes the *difference* far riskier (37, well above both) — hedges only hedge the combination they were designed for. Flip a sign (go long one, short the other, instead of long both) and a risk-reducing pair becomes risk-*amplifying*.

Problem B2 — Why diversification has a floor (the equicorrelation result)

A portfolio holds n assets, each with $\text{Var} = \sigma^2 = 0.04$ and common pairwise correlation $\rho = 0.3$ (so $\text{Cov} = \rho\sigma^2$ for every pair). Find $\text{Var}(\text{average})$ at $n = 50$, and its limit as $n \to \infty$.

Solution.

$$\text{Var}\left(\sum X_i\right) = n\sigma^2 + n(n-1)\rho\sigma^2 = n\sigma^2[1 + (n-1)\rho].$$

$$\text{Var}(\text{average}) = \frac{\text{Var}(\sum X_i)}{n^2} = \frac{\sigma^2[1 + (n-1)\rho]}{n}.$$

At $n = 50$: $\dfrac{0.04[1 + 49(0.3)]}{50} = \boxed{0.01256}$. As $n \to \infty$: $\text{Var}(\text{average}) \to \rho\sigma^2 = \boxed{0.012}$.

★ *Intuition:* This is *the* foundational result behind "systematic risk can't be diversified away." If $\rho = 0$, more assets would drive portfolio variance all the way to zero. But with $\rho = 0.3 > 0$, even infinite diversification ($n \to \infty$) can only shrink variance down to 0.012 — already 95% of the way there at just $n = 50$. The correlation term, not the individual-asset variance, becomes the bottleneck.

Problem B3 — Computing covariance between two indicators from a joint table

$X = \mathbb{I}_A + \mathbb{I}_B$ where $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.15$. Find $\text{Var}(X)$.

Solution.

$$\text{Cov}(\mathbb{I}_A, \mathbb{I}_B) = P(A \cap B) - P(A)P(B) = 0.15 - 0.20 = -0.05.$$

$$\text{Var}(\mathbb{I}_A) = 0.4(0.6) = 0.24, \quad \text{Var}(\mathbb{I}_B) = 0.5(0.5) = 0.25.$$

$$\text{Var}(X) = 0.24 + 0.25 + 2(-0.05) = \boxed{0.39}.$$

★ *Intuition:* This is the bridge from Section I into Section II: the instant you write a random variable as a sum of indicators, "find the covariance" always reduces to "find $P(A \cap B)$ and compare it to $P(A)P(B)$" — a joint-probability lookup, not a new formula. Every variance computed by the indicator method in the rest of this sheet is built from exactly this one move, repeated many times.

II. The Method of Indicator Variables

Problem C1 — The fundamental identity, applied

Prove $\mathbb{E}[\mathbb{I}_A] = P(A)$ directly from the definition of expectation, then use it to find $\mathbb{E}[\#\text{ aces}]$ in a 5-card hand dealt from a 52-card deck (4 aces).

Solution. Proof. $\mathbb{I}_A$ takes value 1 w.p. $P(A)$ and 0 w.p. $1 - P(A)$, so by definition $\mathbb{E}[\mathbb{I}_A] = 1 \cdot P(A) + 0 \cdot (1 - P(A)) = P(A)$. ■ **Application.** Let $\mathbb{I}_j$ = indicator that the jth ace ($j = 1, \ldots, 4$) is among the 5 dealt cards. By symmetry, any specific card lands in the 5-card hand with probability $\frac{5}{52}$:

$$\mathbb{E}[\#\text{aces}] = 4 \cdot \frac{5}{52} = \boxed{\tfrac{5}{13}} \approx 0.385.$$

★ *Intuition:* Notice we never computed the probability of getting exactly 0, 1, 2, 3, or 4 aces separately (which would require a full hypergeometric breakdown) — we completely sidestepped the joint distribution of the hand and just asked "is this one specific ace present?", four times, then added. This sidestep *is* the method.

Problem C2 — Why $\mathbb{I}_A^2 = \mathbb{I}_A$ matters: deriving Bernoulli variance for free

Using only $\mathbb{I}_A \in \{0, 1\}$ (hence $\mathbb{I}_A^2 = \mathbb{I}_A$), prove $\mathrm{Var}(\mathbb{I}_A) = p(1 - p)$ where $p = P(A)$.

Solution.

$$\mathrm{Var}(\mathbb{I}_A) = \mathbb{E}[\mathbb{I}_A^2] - (\mathbb{E}[\mathbb{I}_A])^2 = \mathbb{E}[\mathbb{I}_A] - p^2 \quad (\text{since } \mathbb{I}_A^2 = \mathbb{I}_A)$$

$$= p - p^2 = \boxed{p(1-p)}.$$

★ *Intuition:* The identity $\mathbb{I}_A^2 = \mathbb{I}_A$ is what makes indicator variables so disproportionately powerful: squaring a 0/1 variable never creates anything new ($0^2 = 0$, $1^2 = 1$), so $\mathbb{E}[\mathbb{I}_A^2]$ is always just $\mathbb{E}[\mathbb{I}_A]$ again — no separate second-moment calculation ever required. This single fact is why every variance computation involving indicators in this sheet skips straight from "find p" to "variance is $p(1-p)$," never needing a genuine second-moment integral.

Problem C3 — Indicators indexed by *pairs*, not individuals

100 people are in a room (365 equally likely birthdays). Find $\mathbb{E}[\#\text{ pairs of people who share a birthday}]$.

Solution. Index one indicator per *pair* of people, not per person. There are $\binom{100}{2} = 4950$ pairs, each sharing a birthday with probability $\frac{1}{365}$:

$$\mathbb{E}[X] = \frac{4950}{365} \approx \boxed{13.56} \text{ matching pairs.}$$

★ *Intuition:* This is a different flavor of the same trick: instead of one indicator per "item" (as in C1's aces), we put one indicator on every *pair* of items. Any time a problem asks "how many pairs satisfy some symmetric property," indexing indicators by pairs (giving $\binom{n}{2}$ of them) rather than by individuals is almost always the right move — this is exactly the machinery behind the classic Birthday Pairing result from Notebook 1, derived here in one line instead of a full inclusion–exclusion argument.

Problem D1 — Fixed points in a random permutation: $\mathbb{E}[X] = 1$, always

n people's hats are shuffled into a uniformly random permutation. Let X =# people who get their own hat back. Find $\mathbb{E}[X]$ for general n.

Solution. Let $\mathbb{I}_i = 1$ if person i gets their own hat ($\pi(i) = i$). By symmetry, $\pi(i)$ is equally likely to be any of the n hats, so $P(\mathbb{I}_i = 1) = \frac{1}{n}$.

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[\mathbb{I}_i] = n \cdot \frac{1}{n} = \boxed{1} \quad \text{for } \textit{every } n \geq 1.$$

★ *Intuition:* This is genuinely remarkable: whether 3 people or 3 million people shuffle hats, the *expected* number of correct returns is always exactly 1 — not "approximately 1 for large n," but *exactly* 1, for every single n. The complexity of the full joint distribution of X (which does depend heavily on n, and connects to the derangement numbers D_n from Notebook 1) completely vanishes once you only ask for the mean.

Problem D2 — The harder half: $\mathrm{Var}(X) = 1$, also always

Continuing D1, show $\mathrm{Var}(X) = 1$ for every $n \geq 2$ as well.

Solution. $\mathrm{Var}(\mathbb{I}_i) = \frac{1}{n}\left(1 - \frac{1}{n}\right)$. For $i \neq j$: $\mathbb{E}[\mathbb{I}_i \mathbb{I}_j] = P(\pi(i){=}i, \pi(j){=}j) = \dfrac{1}{n(n-1)}$, so

$$\mathrm{Cov}(\mathbb{I}_i, \mathbb{I}_j) = \frac{1}{n(n-1)} - \frac{1}{n^2} = \frac{1}{n^2(n-1)}.$$

Summing over all $n(n-1)$ ordered pairs $i \neq j$: total covariance contribution $= n(n-1) \cdot \dfrac{1}{n^2(n-1)} = \dfrac{1}{n}$.

$$\mathrm{Var}(X) = n \cdot \frac{1}{n}\left(1 - \frac{1}{n}\right) + \frac{1}{n} = \left(1 - \frac{1}{n}\right) + \frac{1}{n} = \boxed{1}.$$

★ *Intuition:* Mean *and* variance both pinned at exactly 1, for every n — this is one of the most quietly beautiful facts in elementary probability, and it's completely invisible unless you decompose X into indicators. (As $n \to \infty$, the distribution of X converges to Pois(1), which has mean *and* variance both equal to 1 — so this exact finite-n result is really a preview of that limit.)

▶ *Speed trick:* Verified by brute force for $n = 3, 4, 5$ — $\mathrm{Var}(X) = 1$ exactly every time. When a result looks "too clean," that's usually a sign the indicator decomposition is the intended path, not a coincidence to double-check away.

Problem D3 — A beautiful generalization: expected number of 2-cycles

In the same random permutation, let Y = # of 2-cycles (pairs i, j with $\pi(i){=}j$ and $\pi(j){=}i$). Find $\mathbb{E}[Y]$ for general n.

Solution. Let $\mathbb{I}_{ij} = 1$ if $\{i, j\}$ forms a 2-cycle. $P(\pi(i){=}j) = \frac{1}{n}$; given that, $P(\pi(j){=}i) = \frac{1}{n-1}$ (one slot, $n-1$ remaining options), so $P(\mathbb{I}_{ij}{=}1) = \dfrac{1}{n(n-1)}$. There are $\binom{n}{2}$ pairs:

$$\mathbb{E}[Y] = \binom{n}{2} \cdot \frac{1}{n(n-1)} = \frac{n(n-1)/2}{n(n-1)} = \boxed{\tfrac{1}{2}} \quad \text{for } \textit{every } n \geq 2.$$

★ *Intuition:* Another parameter-free constant! This generalizes beautifully: $\mathbb{E}[\#\text{ of } k\text{-cycles}] = \frac{1}{k}$ for *any* fixed k, for all $n \geq k$ (we won't prove the general case here, but the pattern — $\frac{1}{2}$ for 2-cycles, and D1's result is secretly $\frac{1}{1} = 1$ for "1-cycles," i.e. fixed points — should make the general claim believable). The number of cycles of any fixed small length in a random permutation is essentially insensitive to how large the permutation is.

Problem E1 — Expected number of runs in coin flips

A fair coin is flipped $n = 10$ times. A *run* is a maximal block of consecutive identical outcomes. Find $\mathbb{E}[\#\text{ runs}]$.

Solution. Let $R = 1 + \sum_{i=2}^{n} \mathbb{I}_i$, where $\mathbb{I}_i = 1$ if flip i differs from flip $i-1$ (starting a new run); the leading 1 accounts for the first run. For a fair coin, $P(\mathbb{I}_i{=}1) = \frac{1}{2}$ for each of the $n-1$ positions $i = 2, \ldots, n$:

$$\mathbb{E}[R] = 1 + (n-1)\cdot\frac{1}{2} = \frac{n+1}{2} = \boxed{5.5} \quad (n{=}10).$$

★ *Intuition:* "Runs" feels like a global, structural property of the whole sequence, but it decomposes into $n-1$ completely local, easy questions: "does flip i match flip $i-1$?" Summing those local yes/no answers (plus 1 for the very first run) reconstructs the global count exactly — you never need the joint distribution of the full sequence.

Problem E2 — A refinement: counting only head-runs

Same $n = 10$ fair-coin sequence. Find $\mathbb{E}[\#$ of maximal head-blocks] (ignore tail-blocks).

Solution. A head-run starts at position $i = 1$ if flip 1 is H (prob. $\frac{1}{2}$), or at position $i \geq 2$ if flip i is H *and* flip $i{-}1$ is T (prob. $\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$, by independence):

$$\mathbb{E}[H] = \frac{1}{2} + (n-1)\cdot\frac{1}{4} = \frac{1}{2} + \frac{9}{4} = \boxed{2.75}.$$

★ *Intuition:* By the same logic (and symmetry of H/T for a fair coin), $\mathbb{E}[\text{tail-runs}]$ is also exactly 2.75 — and indeed $2.75 + 2.75 = 5.5$, matching E1's total exactly. This cross-check (head-runs + tail-runs = total runs) is a nearly-free sanity check you should always run when a problem splits a count into subcategories: if the pieces don't add back up to the whole, you've made an arithmetic error somewhere.

Problem E3 — Biased coins: runs are maximized exactly at fairness

A coin lands heads w.p. $p = 0.3$. Find $\mathbb{E}[\#$ runs] for $n = 20$ flips, and explain why runs are *most* frequent precisely when $p = \frac{1}{2}$.

Solution. For $i \geq 2$, flip i differs from flip $i{-}1$ with probability $P(H,T) + P(T,H) = 2p(1-p)$:

$$\mathbb{E}[R] = 1 + (n-1)\cdot 2p(1-p) = 1 + 19(0.42) = \boxed{8.98}.$$

★ *Intuition:* $2p(1-p)$ is maximized at $p = \frac{1}{2}$ (giving $\frac{1}{2}$, matching E1's per-step rate exactly) and shrinks toward 0 as $p \to 0$ or $p \to 1$ — which makes total sense: a wildly biased coin produces long, boring, predictable streaks of its favored side (few transitions, few runs), while a perfectly fair coin maximizes genuine unpredictability from one flip to the next. "Maximal disorder/alternation occurs at the symmetric parameter value" is a pattern you'll see across this entire syllabus, from here to the Normal distribution's entropy-maximizing symmetry.

Problem F1 — Expected number of inversions in a random permutation

A random permutation of $n = 10$ elements is drawn. An *inversion* is a pair $i < j$ with $\pi(i) > \pi(j)$. Find $\mathbb{E}[\#$ inversions].

Solution. For each of the $\binom{n}{2}$ pairs $i < j$, by symmetry $P(\pi(i) > \pi(j)) = \frac{1}{2}$ (two random distinct values are equally likely to appear in either order):

$$\mathbb{E}[\#\text{inv}] = \binom{n}{2}\cdot\frac{1}{2} = \frac{n(n-1)}{4} = \boxed{22.5} \quad (n{=}10).$$

★ *Intuition:* "Inversions" measure how far a permutation is from sorted order — it's the exact statistic Kendall's tau and many sorting-algorithm complexity bounds (e.g. insertion sort's running time) are built on. That the expected disorder is exactly *half* the maximum possible disorder is no coincidence, as the next problem makes precise.

Problem F2 — A shuffled deck is exactly 50% disordered, on average

Apply F1's formula to a full 52-card shuffled deck. Compare $\mathbb{E}[\#$inversions] to the maximum possible number of inversions, $\binom{52}{2}$.

Solution.

$$\mathbb{E}[\#\text{inv}] = \frac{52(51)}{4} = \boxed{663}, \qquad \binom{52}{2} = 1326,$$

$$\frac{663}{1326} = \boxed{\tfrac{1}{2}} \text{ exactly.}$$

★ *Intuition:* A freshly riffled deck is, on average, exactly halfway between "perfectly sorted" and "maximally reverse-sorted" — a clean, quotable fact that falls straight out of the $\frac{1}{2}$ per-pair symmetry, with zero extra work beyond F1's formula.

Problem F3 — The hard half: variance of the number of inversions

Find $\text{Var}(\#\text{inversions})$ for $n = 10$. (This requires real care: unlike D2's fixed points, not all pairs-of-indicators behave the same way.)

Solution. Each $\mathbb{I}_{ij} \sim \text{Bernoulli}(\frac{1}{2})$, so $\text{Var}(\mathbb{I}_{ij}) = \frac{1}{4}$, contributing $\binom{n}{2}\cdot\frac{1}{4}$ total. For covariances, three distinct indices $i < j < k$ give rise to 3 pairs of indicators, and (verifiable by direct 6-permutation enumeration on $\{i,j,k\}$) they are *not* all equal:

$$\text{Cov}(\mathbb{I}_{ij}, \mathbb{I}_{ik}) = \text{Cov}(\mathbb{I}_{ik}, \mathbb{I}_{jk}) = +\tfrac{1}{12}$$

(shared index plays the *same* role),

$$\text{Cov}(\mathbb{I}_{ij}, \mathbb{I}_{jk}) = -\tfrac{1}{12}$$

(shared index j plays *opposite* roles — a chain $i{<}j{<}k$).

Pairs of indicators sharing *no* index have covariance exactly 0. Net per triple: $2(\frac{1}{12}) - \frac{1}{12} = \frac{1}{12}$, and there are $\binom{n}{3}$ triples:

$$\text{Var}(X) = \binom{n}{2}\cdot\frac{1}{4} + \binom{n}{3}\cdot\frac{1}{6} = \frac{n(n-1)(2n+5)}{72}.$$

At $n = 10$: $\text{Var}(X) = \frac{10(9)(25)}{72} = \boxed{31.25}$ (sd ≈ 5.59).

★ *Intuition:* This is the chapter's best illustration of why covariance bookkeeping is the genuinely hard part of the indicator method: two inversion-pairs that merely happen to mention a common element can either reinforce each other (+, when the shared element plays matching roles) or partially cancel (-, when it sits "in the middle" of a 3-element chain) — and you cannot tell which without actually checking. Verified here by brute-force enumeration for $n = 3, 4, 5, 6$ before trusting the general formula, exactly as a careful quant should.

Problem G1 — Coupon Collector: the mean, via independent geometric stages

A promotion has $n = 50$ distinct coupon types, drawn uniformly with replacement. Find $\mathbb{E}[T]$, the number of draws until all 50 types are collected.

Solution. Let $G_i = \#$ draws to get a *new* type, immediately after the $(i{-}1)$th distinct type was first obtained. While $i - 1$

types are already owned, each draw is "new" with probability $p_i = \frac{n-i+1}{n}$, so $G_i \sim \text{Geom}(p_i)$, and $T = G_1 + \cdots + G_n$ with the G_i **independent** (memorylessness of each waiting stage):

$$\mathbb{E}[T] = \sum_{i=1}^{n} \mathbb{E}[G_i] = \sum_{i=1}^{n} \frac{n}{n-i+1} = n \sum_{j=1}^{n} \frac{1}{j} = nH_n.$$

At $n = 50$: $H_{50} \approx 4.499$, so $\mathbb{E}[T] = 50(4.499) \approx \boxed{225}$ draws. Euler's approximation: $nH_n \approx n(\ln n + \gamma) = 50(\ln 50 + 0.5772) \approx \boxed{224.5}$ — matches almost exactly.

★ *Intuition:* The *first* coupon is trivial (any draw is new, $p_1 = 1$); the *last* coupon is brutal ($p_{50} = \frac{1}{50}$, expected 50 draws just for the final type). The harmonic sum H_n is precisely "add up how painful each successive stage is" — cheap at the start, catastrophically expensive at the end, which is exactly the lived experience of any real sticker-album or trading-card completion quest.

Problem G2 — Coupon Collector: the variance

Continuing G1 with $n = 50$, find $\text{Var}(T)$ using the independence of the G_i stages.

Solution. Since the G_i are independent, variances add directly (no covariance terms needed — a huge simplification compared to F3!):

$$\text{Var}(T) = \sum_{i=1}^{n} \text{Var}(G_i) = \sum_{i=1}^{n} \frac{1-p_i}{p_i^2}.$$

Substituting $j = n - i + 1$ (so $p = j/n$) and simplifying: $\text{Var}(T) = \sum_{j=1}^{n} \frac{n(n-j)}{j^2}$. At $n = 50$, this evaluates to $\text{Var}(T) \approx \boxed{3838}$, i.e. $\text{sd}(T) \approx \boxed{62}$ draws. Asymptotically, $\text{Var}(T) \sim n^2 \cdot \frac{\pi^2}{6}$ for large n (the famous Basel-problem constant makes a surprise appearance!).

★ *Intuition:* Compare this to F3: there, overlapping indicators forced us into a genuinely intricate covariance accounting. Here, because each stage's waiting time is *independent* of the others (memorylessness strikes again), variances simply add — the decomposition trick doesn't just make the *mean* easy, it makes the *variance* easy too, which is why "break it into independent geometric stages" is such a disproportionately powerful move for any waiting-time problem shaped like this one.

▶ *Speed trick:* Whenever a complicated waiting time can be written as a sum of *independent* simpler waiting times (as opposed to merely identically-distributed-but-dependent indicators, as in F3), always prefer that decomposition — it upgrades "need full covariance bookkeeping" into "just add the variances."

Drill order for review: re-derive D2 (Var of fixed points), F3 (Var of inversions), and G1–G2 (Coupon Collector mean and *variance) from scratch, closed-book. Together they carry the chapter's central tension: when indicators are independent (G), variances just add; when they overlap and interact (D, F), you must carefully classify* which *pairs share structure before summing covariances — confusing these two situations is the most common error at Stage-1 speed.*